

MATRIX / ALLINALL ERP

نقطة تنفيذ n8n — Matrix / AllInAll ERP

٢٥ يونيو ٢٠٢٦

docs/specs/N8N-EXECUTION-PLAN.m

< **المرجع الكامل:** docs/specs/n8n-automation.md < **التوقيت:** بعد اكتمال MVP-1 (Leads + Quotes + Projects + Payments APIs) < **المبدأ:** n8n طبقة أتمتة — **مصدر الحقيقة = Matrix API فقط**

1. متى نبدأ n8n؟

الشروط	الحالة المطلوبة
Events API جاهزة	emit + POST /automation/events من الخدمات
automation_logs table	Prisma migration
Webhook security	HMAC signature + headers
Leads API	GET/PATCH /internal/leads
Automation token	MATRIX_AUTOMATION_TOKEN في n8n

لا تثبت n8n قبل: Auth + Leads CRUD + event emission عند lead.created .

2. البنية التحتية المطلوبة في Matrix (قبل n8n)

2.1 جدول automation_logs

```
model AutomationLog {
  id          String  @id @default(cuid())
  eventName   String
  eventId      String  @unique
  workflowName String?
  status       String  // pending | running | success | failed
  inputPayload Json?
  outputPayload Json?
  errorMessage String?
  retryCount   Int     @default(0)
  startedAt    DateTime @default(now())
  finishedAt   DateTime?
  createdAt    DateTime @default(now())
  updatedAt    DateTime @updatedAt }
```

2.2 جدول domain_events (اختياري — لا idempotency)

```
model DomainEvent {
  id          String  @id @default(cuid())
  eventId      String  @unique
  eventName   String
  payload      Json
  emittedAt   DateTime @default(now())
  processed    Boolean @default(false) }
```

2.3 Event Emitter Service (في apps/api)

كل عملية مهمة تستدعي:

```
await EventEmitter.emit('platform.lead.created', { leadId, leadNumber, ... });
// → يحفظ domain_event // → يرسل webhook لـ n8n مع signature // → يسجل automation_log (status: pending)
```

n8n Headers الصادرة لـ 2.4

```
X-Matrix-Event: platform.lead.created
X-Matrix-Event-Id: evt_abc123 X-Matrix-Signature: hmac-sha256(...) X-Matrix-Timestamp: 1719300000
```

API Endpoints 2.5 (من الوثيقة)

الغرض	Path	Method
استقبال/تسجيل حدث	automation/events/	POST
حالة حدث	automation/events/:id/	GET
n8n أكمل بنجاح	automation/events/:id/complete/	POST
n8n فشل	automation/events/:id/fail/	POST
تسجيل log	automation/logs/	POST
قراءة logs	automation/logs/	GET

3. خريطة الأحداث (Event Catalog)

Platform Events

المرتبطة Workflows	يُصدّر من	Event
New Lead Notification 9.1	LeadService.create	platform.lead.created
Lead Qualified Follow-up 9.3	LeadService.updateStatus	platform.lead.status_changed
Quote Sent 10.1	QuoteService.send	platform.quote.sent
Quote Accepted 10.3	QuoteService.accept	platform.quote.accepted
Customer Onboarding 11.1	CustomerService.create	platform.customer.created
Project Created 12.1	ProjectService.create	platform.project.created
Payment Request Created 13.1	PaymentService.create	platform.payment_request.created
Payment Overdue 13.3	PaymentService (cron/worker)	platform.payment.overdue

ERP Events (مرحلة لاحقة)

Workflows	Event
Low Stock Alert 14.1	erp.inventory.low_stock
Stock Anomaly 14.2	erp.inventory.stock_movement_created
Sales Invoice 15.1	erp.sales.invoice_created
Credit Limit 15.3	erp.sales.order_created
Purchase Approval 16.1	erp.purchase.request_created
Journal Approval 17.1	erp.accounting.journal_entry_created
Leave Request 18.1	erp.hr.leave_requested

ELY Events

Workflows	Event
Knowledge Review 21.1	ely.knowledge.pending_review
Bad Feedback 21.2	ely.feedback.negative

Support Events

Workflows	Event
New Support Request 19.1	support.ticket.created

4. جدول Workflows الكامل (workflow 35)

#	الاسم	Trigger	النوع	المرحلة
9.1	New Lead Notification	<code>platform.lead.created</code>	Event	n8n-1
9.2	Lead Not Contacted Reminder	Cron hourly	Cron	n8n-1
9.3	Lead Qualified Follow-up	<code>platform.lead.status_changed</code>	Event	n8n-1
10.1	Quote Sent	<code>platform.quote.sent</code>	Event	n8n-1
10.2	Quote Follow-up Reminder	Cron daily	Cron	n8n-1
10.3	Quote Accepted	<code>platform.quote.accepted</code>	Event	n8n-2
11.1	New Customer Onboarding	<code>platform.customer.created</code>	Event	n8n-2
11.2	Customer Missing Data	Cron daily	Cron	n8n-2
12.1	Project Created	<code>platform.project.created</code>	Event	n8n-1
12.2	Project Milestone Overdue	Cron daily	Cron	n8n-1
12.3	Weekly Project Report	Cron weekly	Cron	n8n-2
13.1	Payment Request Created	<code>platform.payment_request.created</code>	Event	n8n-1
13.2	Payment Due Soon	Cron daily	Cron	n8n-1
13.3	Payment Overdue	Cron daily	Cron	n8n-1
14.1	Low Stock Alert	<code>exp.inventory.low_stock</code> / Cron 6h	Event+Cron	n8n-2
14.2	Stock Movement Anomaly	<code>exp.inventory.stock_movement_created</code>	Event	n8n-3
15.1	Sales Invoice Created	<code>exp.sales.invoice_created</code>	Event	n8n-2
15.2	Daily Sales Summary	Cron daily 22:00	Cron	n8n-2
15.3	Customer Over Credit Limit	<code>exp.sales.order_created</code>	Event	n8n-3
16.1	Purchase Request Approval	<code>exp.purchase.request_created</code>	Event	n8n-3
16.2	Supplier Invoice Due	Cron daily	Cron	n8n-3
17.1	Journal Entry Approval	<code>exp.accounting.journal_entry_created</code>	Event	n8n-3
17.2	Daily Accounting Check	Cron daily	Cron	n8n-3
18.1	Leave Request Created	<code>exp.hr.leave_requested</code>	Event	n8n-3
18.2	Attendance Missing	Cron daily	Cron	n8n-3

المرحلة	النوع	Trigger	الاسم	#
n8n-3	Cron	Cron monthly	Payroll Ready	18.3
n8n-2	Event	<code>support.ticket.created</code>	New Support Request	19.1
n8n-2	Cron	Cron hourly	Support SLA Breach	19.2
n8n-2	Cron	Cron daily	Daily Owner Report	20.1
n8n-2	Cron	Cron weekly	Weekly Business Report	20.2
n8n-3	Event	<code>ely.knowledge.pending_review</code>	Knowledge Needs Review	21.1
n8n-3	Event	<code>ely.feedback.negative</code>	ELY Bad Feedback	21.2
n8n-3	Cron	Cron weekly	Weekly ELY Quality Report	21.3
n8n-3	Cron	Cron daily	Database Backup	22.1
n8n-3	Cron	Cron daily	Files Backup	22.2
n8n-1	Cron	Cron 5min	API Health Check	23.1
n8n-1	Internal	n8n error trigger	Failed Workflow Alert	23.2

5. مراحل تنفيذ n8n (4 مراحل)

المرحلة n8n-1 – الأساس + Platform (أسبوع 1 بعد MVP-1)

البنية:

n8n server (n8n cloud أو VPS على Docker) ☐

`MATRIX_AUTOMATION_TOKEN` + `MATRIX_WEBHOOK_SECRET` ☐

Automation logs API ☐

Event emitter في Lead/Quote/Project/Payment services ☐

Webhook signature verification node (n8n Function) ☐

:Workflows (7)

1. New Lead Notification (9.1)

2. Lead Not Contacted Reminder (9.2)

3. Quote Sent (10.1)

4. Quote Follow-up Reminder (10.2)

5. Payment Due Soon (13.2)

6. Project Milestone Overdue (12.2)

7. API Health Check (23.1) + Failed Workflow Alert (23.2)

قنوات الإشعار في هذه المرحلة: Email (SMTP) فقط — WhatsApp لاحقًا.

المرحلة n8n-2 — توسع Platform + ERP خفيف (أسبوع 2-3)

:Workflows (10)

1. Quote Accepted (10.3) — يستدعي convert customer + create project عبر API
2. Customer Onboarding (11.1) + Missing Data (11.2)
3. Project Created (12.1) + Weekly Report (12.3)
4. Payment Request Created (13.1) + Payment Overdue (13.3)
5. Low Stock Alert (14.1)
6. Daily Sales Summary (15.2)
7. Support workflows (19.1, 19.2)
8. Daily Owner Report (20.1) + Weekly Business (20.2)

المرحلة n8n-3 — كامل HR + ELY + Backups

:Workflows (15) باقي ERP + HR + Accounting + ELY + Backups

المرحلة n8n-4 — تكاملات متقدمة

1. WhatsApp API
2. Email templates احترافية
3. Multi-tenant automation settings
4. Google Sheets sync
5. Advanced branching

6. قالب Workflow موحد (كل workflow يتبعه)

1. Trigger (Webhook / Cron)

1. Idempotency Check

→ GET /automation/events/:eventId OR check automation_logs
→ if already processed → STOP

1. Validate signature (webhooks only)

2. Main Logic (HTTP Request to Matrix API / External)

3. On Success

→ POST /automation/logs (status: success)
→ POST /automation/events/:id/complete

1. On Error

→ Retry (max 3, exponential backoff)
→ Fallback (Email if WhatsApp fails) → POST /automation/events/:id/fail → Notify internal (Telegram/Email)

7. ترتيب بناء API لدعم n8n

الترتيب	API	يخدم Workflows
N1	Automation module (logs, events)	الكل
N2	<code>=GET /internal/leads?status=&olderThan</code>	9.2
N3	<code>POST /internal/leads/:id/notes</code>	9.3
N4	<code>POST /internal/quotes/:id/send</code>	10.1
N5	<code>GET /internal/quotes?status=sent&noResponseSince=48h</code>	10.2
N6	Quote accept → customer + project APIs	10.3
N7	<code>GET /internal/projects/overdue</code>	12.2
N8	<code>POST /internal/projects/:id/tasks</code>	12.1
N9	Payment status PATCH	13.3
N10	<code>GET /erp/inventory/low-stock</code>	14.1
N11	<code>GET /erp/sales/daily-summary</code>	15.2
N12	<code>GET /erp/accounting/daily-check</code>	17.2
N13	<code>GET /erp/hr/attendance-missing</code>	18.2

8. مهام Matrix API قبل كل workflow (micro-tasks)

TASK-N8N-001: Automation module skeleton

1. Prisma: `AutomationLog`, `DomainEvent`
2. `POST/GET /automation/logs`
3. Build + test

TASK-N8N-002: Event emitter service

1. `EventEmitterService.emit(name, payload)`
2. HMAC signature generation
3. Webhook POST to `N8N_WEBHOOK_URL`

TASK-N8N-003: Wire `platform.lead.created`

1. في `LeadService.create()` بعد الحفظ
2. اختبار: إنشاء lead → يظهر في `automation_logs`

TASK-N8N-004: n8n Docker + workflow 9.1

docker-compose service: n8n .1

/infra/n8n/workflows في workflow JSON export .2

end-to-end اختبار .3

TASK-N8N-005 → N8N-035: workflow واحد لكل مهمة

قاعدة: workflow واحد = مهمة واحدة = اختبار واحد.

9. هيكل الملفات المقترح

```
infra/
├── n8n/ | ├── docker-compose.n8n.yml | ├── .env.example | └── workflows/ | └── 09-01-new-lead-notification.json | └── 09-02-lead-not-contacted.json | └── ... apps/api/src/ ├── automation/ | ├── automation.module.ts | ├── automation.controller.ts | ├── automation-log.service.ts | ├── event-emitter.service.ts | └── webhook-signature.service.ts
```

10. Environment Variables

Matrix API (.env)

```
N8N_WEBHOOK_BASE_URL=https://n8n.example.com/webhook
N8N_WEBHOOK_SECRET=... MATRIX_AUTOMATION_TOKEN=... # API يستدعي n8n للـ
```

n8n (.env)

```
MATRIX_API_BASE_URL=https://api.matrix.example.com
MATRIX_AUTOMATION_TOKEN=... MATRIX_WEBHOOK_SECRET=... SMTP_HOST=... SMTP_PORT=... SMTP_USER=...
SMTP_PASS=...
```

11. Error Handling الموحّد

```
WhatsApp فشل → retry ×2 → Email → internal task → log failure
API timeout → retry ×3 (5s, 15s, 45s) → fail event → alert Duplicate eventId → skip silently → log "skipped_duplicate"
```

12. Idempotency — التنفيذ

1. Matrix يولّد `eventId` فريد (cuid) لكل حدث
2. n8n أول `node: HTTP GET /automation/logs?eventId=X`
3. إذا `status=success` موجود → Stop workflow
4. كل callback يحمل نفس `eventId`

13. الربط مع خطة المشروع الرئيسية

(أسابيع 1-4) MVP-1

→ Leads, Quotes, Projects, Payments APIs → TASK-N8N-001 إلى N8N-003

أساسية n8n server + 7 workflows (أسبوع 5) n8n-1

ERP events emission → n8n-2 + n8n-3 (أسابيع 6-10) MVP-2 ERP

n8n-3 ELY workflows → n8n-4 integrations MVP-3 + ELY

14. Definition of Done — لكل Workflow

- ☐ Trigger واضح وموثّق
- ☐ Idempotency check
- ☐ Signature validation (إن webhook)
- ☐ Retry + error branch
- ☐ عند النجاح/الفشل `POST /automation/logs`
- ☐ لا secrets في workflow JSON (فقط credentials)
- ☐ اختبار يدوي + تسجيل في sprint-log
- ☐ JSON محفوظ في `/infra/n8n/workflows`

15. قواعد للمطور / Agent

1. لا تبني 35 workflow دفعة واحدة — واحد لكل TASK
2. لا تعدّل DB من Matrix API — n8n فقط
3. كل تعديل في Matrix يرسل event قبل بناء workflow يعتمد عليه
4. اختبر بدون WhatsApp أولاً — Email fallback
5. صدّر workflow JSON إلى Git للمراجعة

آخر تحديث: 25-06-2025 — مدمج من وثيقة n8n الكاملة

docs/specs/N8N-EXECUTION-PLAN.md — ٢٥ يونيو ٢٠٢٦ — Matrix Platform